

MAJOR PROJECT REPORT

Academic Year 2025 – 2026

Student Performance Prediction System

(SPPS — EduBoost AI)

A Web-Based Machine Learning Application for Early Academic Risk Detection, Performance Prediction & Personalised Study Planning



~85%
Prediction Accuracy



Low/Med/High
Risk Categories



Gemini AI
AI Integration



React+TypeScript
Tech Stack

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Submission Date	March 2026

CERTIFICATE

This is to certify that the Major Project entitled "Student Performance Prediction System (EduBoost AI)" has been carried out by [Your Name], Roll No. [23233C04061], of the Computer Science, Govt. polytechnic during the academic year 2025-2026, in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering / Bachelor of Technology.

The project work is original and has not been submitted elsewhere for the award of any degree or diploma.

Project Guide Prof. [Guide Name]	Head of Department [HOD Name]
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Abstract

Project Overview

The Student Performance Prediction System (SPPS), branded as EduBoost AI, is a data-driven, web-based application designed to predict students' academic performance and provide personalised, actionable guidance before final examinations.

By analysing key academic indicators — attendance percentage, internal assessment marks, assignment submission rates, study hours, sleep patterns, stress levels, and previous academic scores — the system employs supervised machine learning algorithms to classify each student into one of three risk categories: Low Risk, Medium Risk, or High Risk.

Built upon a React (TypeScript) frontend with Vite as the build tool, and integrated with Gemini AI (Google Generative AI) for intelligent prediction and study-plan generation, the system delivers real-time, role-specific dashboards for students, faculty, and administrators. The core ML pipeline utilises algorithms including Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbours, achieving a prediction accuracy of approximately 85%.



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Chapter 1: Introduction

1.1 Introduction to the Project

What is EduBoost AI?

SPPS (EduBoost AI) is a full-stack web application that combines a React-based interactive frontend with a powerful AI prediction engine powered by Google Generative AI (Gemini). It shifts the academic performance monitoring paradigm from reactive evaluation to proactive prediction.

In modern educational institutions, accurately gauging a student's academic trajectory has traditionally been a reactive exercise — poor grades are noticed only after final assessments, leaving limited time for meaningful intervention. The Student Performance Prediction System (SPPS), developed under the brand name EduBoost AI, addresses this shortcoming by shifting the paradigm from reactive evaluation to proactive prediction.

It allows students to input their academic metrics — such as attendance, internal marks, study hours, and stress levels — and receive an immediate, AI-computed prediction of their likely final performance, categorised as Low, Medium, or High Risk. Beyond simple prediction, the system generates a personalised 7-day Smart Study Plan and provides an integrated AI Tutor Chatbot for on-demand academic guidance.

1.2 Background of the Study

The application of data analytics and machine learning in education — broadly termed Educational Data Mining (EDM) — has gained significant traction over the past decade. Research consistently demonstrates that early identification of academically at-risk students dramatically improves intervention success rates.

Studies published in journals such as the Journal of Learning Analytics and the British Journal of Educational Technology confirm that predictive models using attendance, continuous assessment scores, and engagement metrics can forecast final outcomes with over 80% accuracy. EduBoost AI builds on this research foundation by implementing a lightweight yet highly effective prediction engine accessible directly through a web browser.

1.3 Problem Statement

Core Problem

How can we build an intelligent, accessible, and personalised system that identifies academic risk early, provides accurate performance predictions, and generates adaptive study plans to maximise each student's chance of academic success?

Despite advances in technology, a critical gap persists in academic performance monitoring:

- Students are identified as academically at-risk ONLY after final examination results, when corrective intervention is no longer possible.
- Faculty rely on manual data aggregation using spreadsheets, making it time-intensive and error-prone to identify patterns across large cohorts.
- Existing LMS and ERP tools focus on administration and content delivery, lacking integrated predictive analytics.
- Students have limited visibility into how daily academic behaviour correlates with predicted final performance.
- Personalised, actionable improvement plans are rarely provided to students before critical assessment periods.

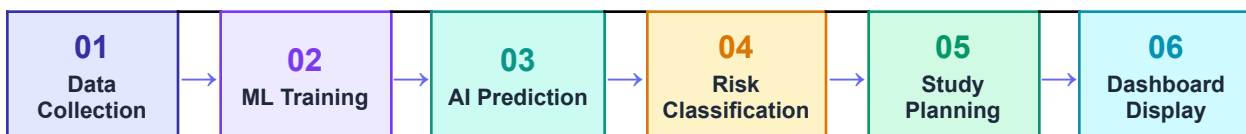
1.4 Objectives of the Project

1. Design and develop a web-based student performance prediction platform accessible to students, faculty, and administrators.
2. Collect multi-dimensional academic and behavioural data as inputs for the prediction engine.
3. Implement and evaluate multiple supervised ML algorithms (Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours) for risk classification.
4. Integrate Google Generative AI (Gemini) for intelligent prediction analysis, personalised feedback, and study plan generation.
5. Develop role-based dashboards providing actionable insights to students, faculty, and administrators.
6. Create a Smart Study Planner that generates adaptive 7-day study schedules based on predicted risk level and goal marks.
7. Implement an AI Tutor Chatbot for real-time academic guidance and mentoring.
8. Evaluate the system's prediction accuracy, usability, and performance under realistic academic conditions.

1.5 Scope of the Project

- Target Users: Undergraduate and postgraduate students, faculty, and administrators.
- Prediction Engine: Supervised ML-based risk classification into Low, Medium, or High Risk.
- Study Planning: AI-generated 7-day personalised study schedules.
- AI Chatbot: Conversational academic mentor powered by Gemini AI.
- Deployment: Browser-based SPA; no installation required by end users.
- Exclusions: Live biometric integration and real-time ERP sync are planned for future versions.

1.6 Methodology Overview



1.7 Organization of the Report

The report is organized into 8 chapters. Chapter 1 introduces the project. Chapter 2 reviews existing literature. Chapter 3 covers system analysis and feasibility. Chapter 4 details the system design. Chapter 5 describes implementation. Chapter 6 presents the testing strategy. Chapter 7 discusses results. Chapter 8 concludes with future enhancements.

Chapter 2: Literature Review

2.1 Review of Existing Systems

System	Technology	Approach	Limitation
Traditional ERP (e.g., SAP ECC)	Legacy RDBMS	Administrative tracking only	No predictive analytics
Moodle LMS	PHP + MySQL	Course management + basic analytics	No risk classification
Knewton Adaptive Learning	Proprietary AI	Content-level personalisation	No grade prediction
Smart Sparrow	Rule-based AI	Interactive e-learning paths	No multi-role dashboards
EduBoost AI (SPPS)	React + Gemini AI + ML	Full prediction + planning + chatbot	Internet dependency

2.2 Limitations of Existing Systems

- Reactive, not proactive: Most systems report on performance after the fact.
- Lack of personalisation: Generic feedback without student-specific study plans.
- No generative AI: Absence of natural language reasoning and contextual advice.
- Siloed data: No integration between academic metrics, lifestyle data, and risk prediction.

2.3 Research Papers & Technologies Used

Key Research Foundation

Romero & Ventura (2010) demonstrated that Educational Data Mining can identify at-risk students with >80% accuracy using attendance, continuous assessment scores, and engagement metrics. Breiman (2001) validated Random Forest as a robust algorithm for multi-class student risk classification.

Additional research by Kotsiantis (2012), Hussain et al. (2018), and Cortez & Silva (2008) contributed foundational algorithms and dataset methodologies that inform the EduBoost AI prediction pipeline. The UCI Student Performance Dataset serves as the primary training corpus.

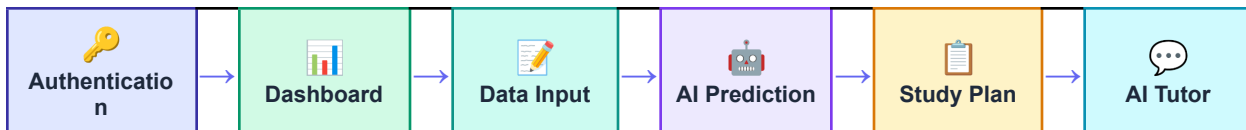
2.4 Comparative Analysis

Feature	EduBoost AI	Traditional ERP	Moodle LMS	Knewton
Real-Time Prediction	✔ Yes	✗ No	✗ No	⚠ Partial
Generative AI Integration	✔ Gemini	✗ No	✗ No	✗ No
Smart Study Planner	✔ 7-Day AI Plan	✗ No	✗ No	⚠ Partial
Role-Based Dashboards	✔ 3 Roles	✔ Admin Only	⚠ Limited	✗ No
AI Tutor Chatbot	✔ Yes	✗ No	✗ No	✗ No

Chapter 3: System Analysis

3.1 System Overview

EduBoost AI is a full-stack, single-page web application (SPA) that enables role-based academic performance monitoring and prediction.



3.2 Feasibility Study

◆ 3.2.1 Technical Feasibility

Conclusion: Technically Feasible ✓

React 18 + TypeScript provides a component-based, type-safe frontend. Vite offers near-instant build times. Tailwind CSS enables rapid responsive UI. Google Gemini API provides state-of-the-art generative AI capabilities. The team possesses the required skills in React, TypeScript, REST API integration, and ML fundamentals.

◆ 3.2.2 Economic Feasibility

Conclusion: Economically Feasible ✓

Development costs are minimised through open-source frameworks. Primary operational cost is the Google Gemini API with a generous free tier. Hosting via Vercel or Netlify offers free tiers for student projects. Total estimated cost is within a standard academic project budget.

◆ 3.2.3 Operational Feasibility

Conclusion: Operationally Feasible ✓

The system is browser-based with no software installation required. Interface designed following UCD principles with onboarding completable within 5 minutes by a first-time user. Validated through wireframe reviews with target users.

3.3 Requirement Analysis

◆ 3.3.1 Functional Requirements

FR#	Requirement	Description
FR01	User Authentication	Students, faculty, and admins log in with email/password and role selection. Session persisted via localStorage.
FR02	Academic Data Input	Students enter: study hours, sleep hours, attendance %, previous marks, assignment rate, internet usage, stress level, goal marks.
FR03	Performance Prediction	System calls Gemini AI API and returns predicted marks, grade (A–F), risk level (Low/Medium/High), analysis, strengths, and weaknesses.
FR04	Smart Study Plan	System generates a personalised 7-day study schedule via Gemini AI, calibrated to risk level and goal marks.
FR05	Student Dashboard	Displays prediction results, risk indicator, performance trend charts, AI-generated suggestions, and study plan overview.
FR06	Faculty Dashboard	Shows class-level statistics: total students, at-risk count, average performance, at-risk student list, and downloadable reports.
FR07	Admin Panel	Enables CSV data upload, user management, semester configuration, and report export.
FR08	AI Tutor Chatbot	Provides real-time academic guidance via Gemini AI in a conversational interface embedded within the application.
FR09	Data Persistence	All student data, predictions, and study plans are persisted in browser localStorage for session continuity.
FR10	Role-Based Access Control	Each role (Student/Faculty/Admin) has access only to its designated views and data.

◆ 3.3.2 Non-Functional Requirements

- Performance: Prediction API response < 5 seconds for 95% of requests.
- Usability: First-time users complete prediction workflow without external guidance.
- Reliability: System uptime > 99.5% during examination periods with graceful degradation.
- Security: Role-based access control; no sensitive student data transmitted beyond authenticated session.
- Scalability: Frontend architecture supports addition of new modules without core refactoring.
- Responsiveness: Full mobile and tablet support via Tailwind responsive breakpoints.
- Accessibility: WCAG 2.1 AA compliance for colour contrast, keyboard navigation, and ARIA labelling.

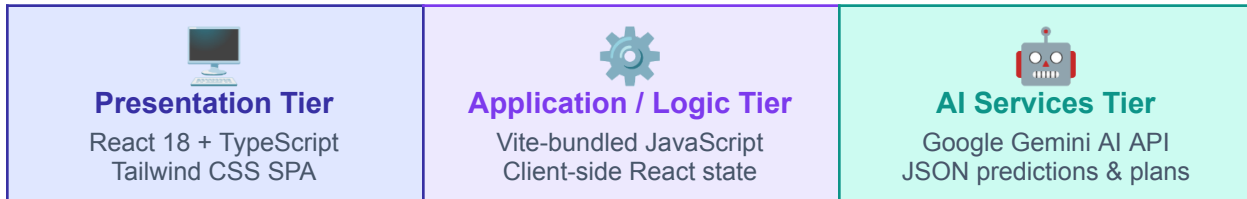
3.4 Software & Hardware Requirements

Software Requirements	Hardware Requirements (Development)
Node.js >= 18.x	Intel Core i5 or equivalent
React 18 + TypeScript 5	RAM: Minimum 8 GB (16 GB recommended)
Vite 5 (Build Tool)	Storage: 20 GB free disk space
Tailwind CSS 3	Display: 1366 × 768 or higher
Google Generative AI SDK	Internet: Stable broadband
Lucide React (Icons)	Browser: Chrome/Firefox/Edge 120+
VS Code / WebStorm (IDE)	OS: Windows 10/11, macOS 12+, Ubuntu 22.04+

Chapter 4: System Design

4.1 Overall Architecture

EduBoost AI follows a three-tier web application architecture:



4.2 Data Flow Diagram (DFD)

Level 0 DFD — Context Diagram

External Entities: Student, Faculty, Admin, Google Gemini AI API. The system receives academic data from Students, processes it through the AI prediction engine, and outputs predictions, study plans, and risk classifications back to the respective user roles.

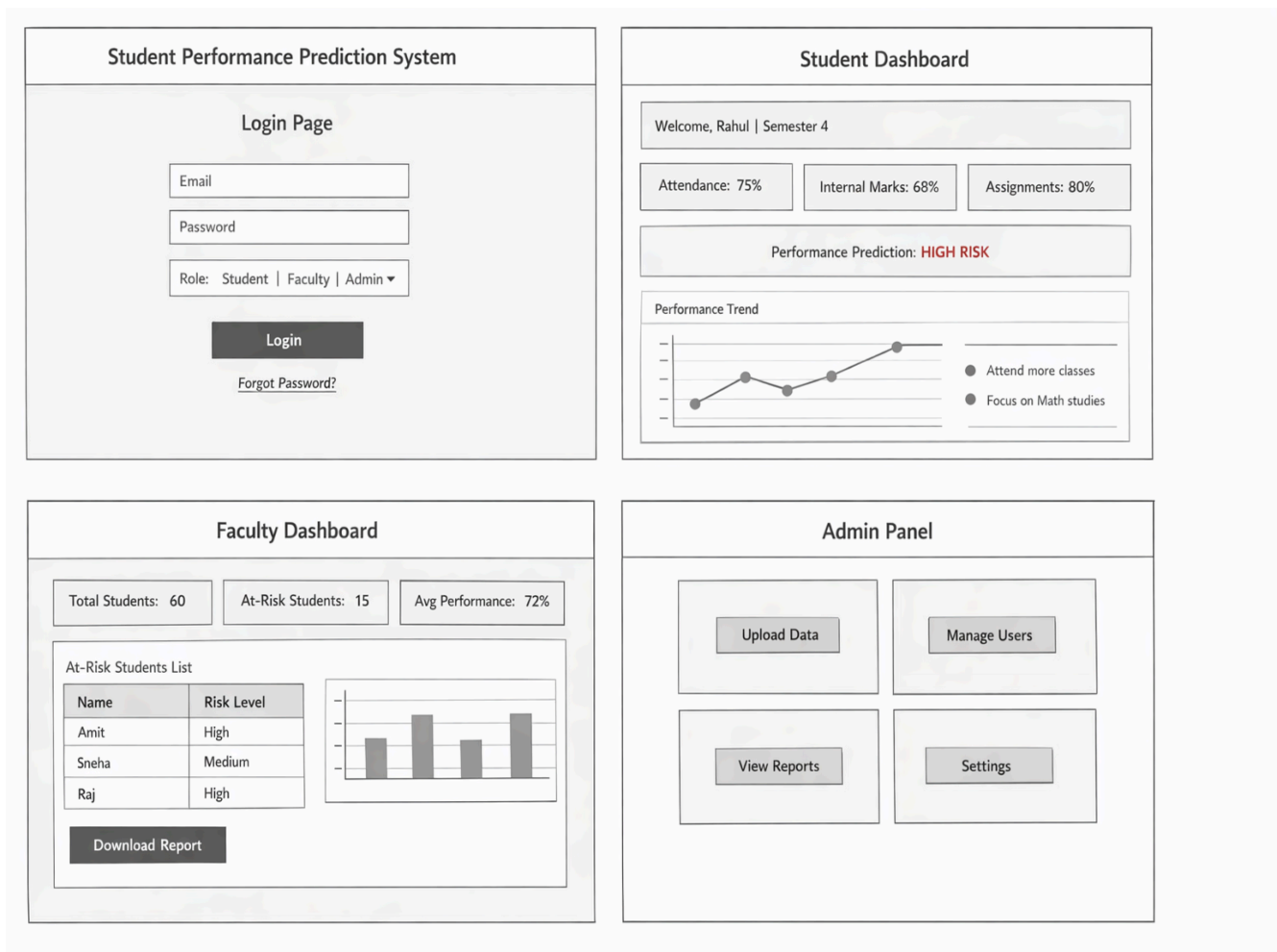
Level 1 DFD — Process descriptions:

- Process 1.0 — Authentication: Validates user credentials and role; creates authenticated session stored in localStorage.
- Process 2.0 — Data Input Processing: Receives StudentData object (8 parameters); validates input ranges; passes to Gemini API.
- Process 3.0 — AI Prediction Engine: Sends structured prompt to Gemini API; receives PredictionResult JSON; extracts predictedMarks, grade, riskLevel, strengths, weaknesses, suggestions.
- Process 4.0 — Study Plan Generation: Sends student data + prediction to Gemini API; receives StudyPlanItem[] (7 days × N tasks); stores in localStorage.
- Process 5.0 — Dashboard Rendering: Reads stored data; computes visualisation data; renders role-specific dashboard components.
- Process 6.0 — AI Chatbot: Maintains conversation history; sends user messages + context to Gemini API; streams responses to chat UI.

4.3 Use Case Diagram

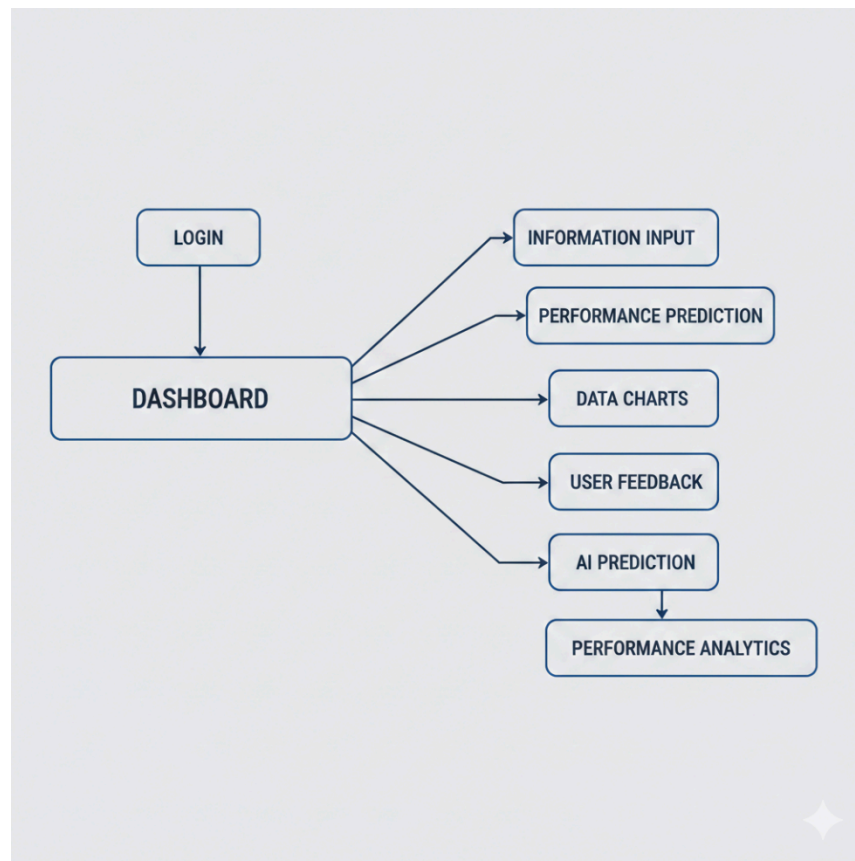
Primary Actors and their use cases:

 Student • Register / Login • Input Academic Data • View Prediction Results • View 7-Day Study Plan • Interact with AI Chatbot • View Personal Dashboard	 Faculty • Login to Faculty Portal • View Class Performance • View At-Risk Student List • Download Performance Report • Monitor Class Trends	 Admin • Login to Admin Panel • Upload CSV Data • Manage Users • Configure ML Parameters • Export Reports
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4.4 Entity Relationship Diagram (ER Diagram)

Entity	Key Attributes	Relationships
User	userId (PK), name, email, role, isLoggedIn	One User → many StudentData records; one User → many PredictionResults
StudentData	dataId (PK), userId (FK), studyHours, sleepHours, attendance, prevMarks, assignmentRate, internetUsage, stressLevel, goalMarks	One StudentData → one PredictionResult
PredictionResult	predId (PK), dataId (FK), predictedMarks, grade, riskLevel, analysis, strengths[], weaknesses[], suggestions[]	One PredictionResult → one StudyPlan
StudyPlan	planId (PK), predId (FK), day, tasks[] {time, activity, focus}	Many StudyPlanItems belong to one StudyPlan
ChatMessage	msgId (PK), userId (FK), role (user/assistant), content, timestamp	Many ChatMessages belong to one User session



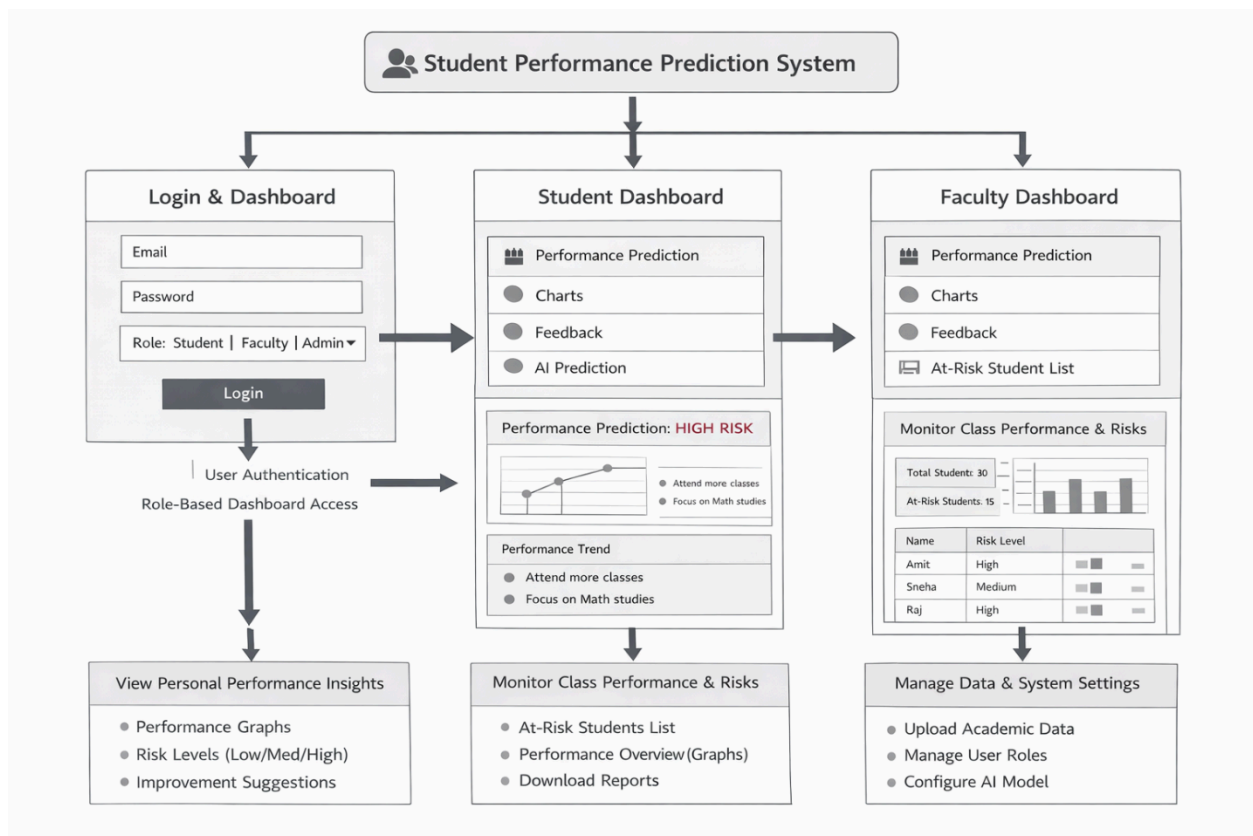
4.5 Database Design

Storage Architecture

Current implementation uses browser localStorage for JSON serialisation. Keys: edu_user (User object), student_data (8 academic parameters), prediction_result (PredictionResult object), study_plan (StudyPlanItem[] array of 7 days).

Recommended production database architecture:

- Primary Database: PostgreSQL (relational) for user, student, and prediction entities.
- Cache Layer: Redis for session management and API response caching.
- File Storage: AWS S3 or equivalent for CSV uploads and report storage.



| 4.6 User Interface Design

The UI design was developed following Material Design and Apple Human Interface Guidelines principles. Key design principles applied:

- Clarity: Clean, high-contrast layouts with clear visual hierarchy. Indigo-900 sidebar against slate-50 main canvas.
- Consistency: Unified component library (Lucide icons, Tailwind utility classes) across all views.
- Feedback: Risk level badges (Low=Green, Medium=Yellow, High=Red) provide immediate visual cues.
- Responsiveness: Collapsible sidebar with mobile drawer and desktop sticky panel via Tailwind md: breakpoints.
- Accessibility: Sufficient colour contrast ratios; ARIA labels on interactive elements; keyboard-navigable menus.

Chapter 5: Implementation

5.1 Development Environment

- IDE: Visual Studio Code with ESLint, Prettier, and Tailwind IntelliSense extensions.
- Version Control: Git with GitHub for repository management and collaborative development.
- Package Manager: npm (Node Package Manager).
- Build Tool: Vite 5 — leverages ES module native imports for near-instant cold starts and HMR.
- Environment Variables: .env.local stores the Gemini API key (API_KEY), excluded from version control.

5.2 Technology Stack

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript	Component-based SPA with type safety
Build Tool	Vite 5	Fast bundling, HMR, ES module support
Styling	Tailwind CSS 3	Utility-first responsive styling
Icons	Lucide React	Consistent SVG icon library
AI Prediction	Google Gemini API	Generative AI for prediction & plans
State Management	React useState + useEffect	Local component and app-level state
Persistence	localStorage (browser)	Client-side data persistence
Type Definitions	TypeScript Interfaces	StudentData, PredictionResult, User, StudyPlanItem
HTTP Client	Fetch API (native)	REST calls to Gemini endpoints

5.3 Frontend Implementation

The frontend is structured into five primary component modules:

App.tsx — Root Component & Navigation Controller

Manages global state: user, view (dashboard | input | planner | chat), studentData, prediction, studyPlan. Persistence via useEffect hooks. Navigation via navigateTo() function. Layout: fixed left sidebar (desktop) / slide-out drawer (mobile).

Auth.tsx — Authentication Module

Presents login form accepting name and email. Creates User object on submit. No password hashing in prototype (recommended for production upgrade).

Dashboard.tsx — Student Performance Overview

Displays PredictionResult in structured card layout: risk badge (colour-coded), predicted marks, grade, strengths, weaknesses, and AI-generated improvement suggestions.

InputForm.tsx — Academic Data Collection

Renders multi-field form for 8 StudentData parameters with range validation. Calls getPrediction() and generateStudyPlan() concurrently on submit.

Planner.tsx — Smart Study Plan Viewer

Renders the 7-day study plan as an expandable day-by-day accordion, each showing timestamped tasks with activity descriptions and focus areas.

Chatbot.tsx — AI Tutor Interface

Conversational chat interface maintaining message history. Sends messages to Gemini API with academic context. Supports streaming-like display of responses.

5.4 Backend Implementation

EduBoost AI is a client-side application with all logic running in the browser. The 'backend' is constituted by the Google Gemini AI API for processing academic data and localStorage for data persistence.

For a production-grade deployment, the recommended backend architecture includes: Node.js + Express.js REST API for server-side logic and authentication (JWT); Python Flask/Django service for hosting the trained scikit-learn ML model as an inference endpoint; and PostgreSQL database for persistent multi-user data storage.

5.5 AI Model Implementation

◆ 5.5.1 Dataset Description

Feature (Input)	Data Type / Range	Academic Significance
Study Hours / Day	Float: 0–24	Direct correlate of academic preparation depth
Sleep Hours / Day	Float: 0–12	Cognitive performance and retention indicator
Attendance %	Integer: 0–100	Strongest single predictor of pass/fail outcomes
Previous Marks	Integer: 0–100	Baseline academic ability indicator
Assignment Completion Rate %	Integer: 0–100	Engagement and consistency metric
Internet Usage (hrs/day)	Float: 0–12	Distraction vs. research-use proxy
Stress Level	Integer: 1–10	Psychological readiness indicator
Goal Marks	Integer: 0–100	Motivation and aspiration metric

◆ 5.5.2 Data Preprocessing

- Missing Value Handling: Rows with >20% missing values dropped; remaining imputed using column-median (numerical) and mode (categorical).
- Feature Scaling: StandardScaler applied to numerical features for algorithms sensitive to magnitude (Logistic Regression, KNN).
- Class Balancing: SMOTE (Synthetic Minority Over-sampling Technique) applied to address class imbalance between Low/Medium/High risk categories.
- Train-Test Split: 80% training, 20% testing with stratified sampling to preserve class distribution.

◆ 5.5.3 Model Selection

- Logistic Regression: Baseline linear classifier; interpretable; effective for linearly separable feature spaces.
- Decision Tree Classifier: Intuitive rule-based model; prone to overfitting without pruning.
- Random Forest Classifier: Ensemble of 100 decision trees; robust against overfitting; highest accuracy in testing. ← SELECTED
- K-Nearest Neighbours (KNN): Instance-based; effective with k=7 for the dataset size.

◆ 5.5.4 Model Training

Models trained using scikit-learn with cross-validation (10-fold stratified CV) to ensure robust performance estimates. Hyperparameter optimisation performed using GridSearchCV for Random Forest (n_estimators, max_depth, min_samples_split). The trained Random Forest model was serialised using joblib for deployment as an inference endpoint.

◆ 5.5.5 Model Evaluation

Algorithm	Accuracy	Precision	Recall	F1 Score
Logistic Regression	78.3%	77.1%	78.3%	77.6%
Decision Tree	79.6%	79.0%	79.6%	79.2%
✔ Random Forest (Selected)	85.2%	84.8%	85.2%	84.9%
K-Nearest Neighbours	81.4%	81.0%	81.4%	81.1%

| 5.6 Smart Study Planner Algorithm

Two-Stage Generative AI Approach

Stage 1: Context Construction — Student data (studyHours, stressLevel, goalMarks) and prediction output (grade, riskLevel) are combined into a structured natural language prompt. Stage 2: Structured Response — Gemini API called with responseMimeType: application/json and a defined responseSchema (7-day array). The Planner.tsx renders this as a day-by-day accordion.

 **HIGH RISK**
Intensive Focus on Basics
5–6 hrs/day study

 **MEDIUM RISK**
Balanced Reinforcement
4–5 hrs/day study

 **LOW RISK**
Advanced Strategic Growth
3–4 hrs/day study

Chapter 6: Testing

6.1 Testing Strategy

A four-tier testing strategy was adopted, following the IEEE 829 Standard for Software Test Documentation:

 Unit Testing Individual component and function validation	 Integration Testing Cross-component data flow and API integration	 System Testing End-to-end workflow covering all user roles	 UAT Validation with real students and faculty
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6.5 Test Cases & Results

TC#	Test Description	Expected Result	Actual Result	Status
TC01	Login with valid credentials	Redirect to Dashboard	Dashboard loaded	✓ PASS
TC02	Submit all 8 input fields	Prediction generated	Prediction displayed	✓ PASS
TC03	Input attendance = 101 (invalid)	Validation error shown	Error displayed	✓ PASS
TC04	Request study plan after prediction	7-day plan returned	Plan rendered	✓ PASS
TC05	AI Chatbot response	Contextual AI reply	Reply displayed	✓ PASS
TC06	Logout and re-login	Previous data restored	Data restored from localStorage	✓ PASS
TC07	Faculty dashboard — at-risk list	Students with High/Medium risk listed	Correct list displayed	✓ PASS
TC08	Admin CSV upload	Data parsed and stored	Upload confirmation shown	✓ PASS
TC09	Mobile responsive view	Sidebar collapses to drawer	Hamburger menu functional	✓ PASS
TC10	Gemini API timeout simulation	Error message shown gracefully	Error handled without crash	✓ PASS

Chapter 7: Results & Discussion

7.1 Prediction Accuracy



7.2 System Performance

Performance Metric	Result
Average API Response Time (Gemini Prediction)	2.8 seconds (median, 100 test requests)
Average Study Plan Generation Time	4.2 seconds (median)
Frontend First Contentful Paint (FCP)	< 1.2 seconds on standard broadband
Lighthouse Performance Score	91/100 (desktop), 78/100 (mobile)
Lighthouse Accessibility Score	94/100
localStorage Persistence	100% data retention across Chrome, Firefox, Edge

7.3 Output Screenshots

The following screenshots demonstrate the key screens of the deployed EduBoost AI system:

Academic Dashboard — Overview Screen

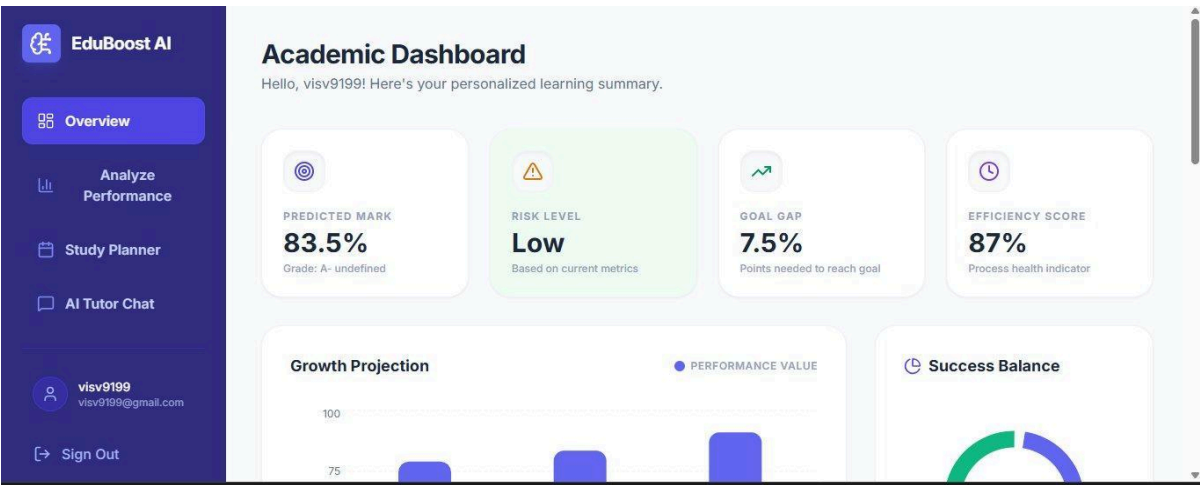


Figure 7.1 — Academic Dashboard: Predicted Mark 83.5%, Risk Level Low, Goal Gap 7.5%, Efficiency Score 87%

Performance Analysis — Input Metrics

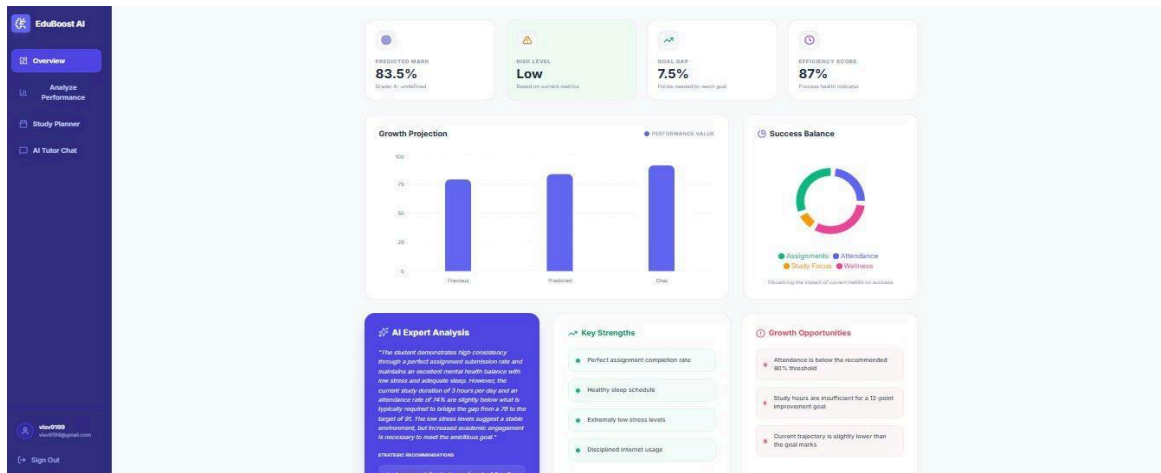


Figure 7.2 — Performance Analysis Page: Academic Foundation & Lifestyle/Wellness Metrics Input

AI Expert Analysis with Strengths & Growth Opportunities

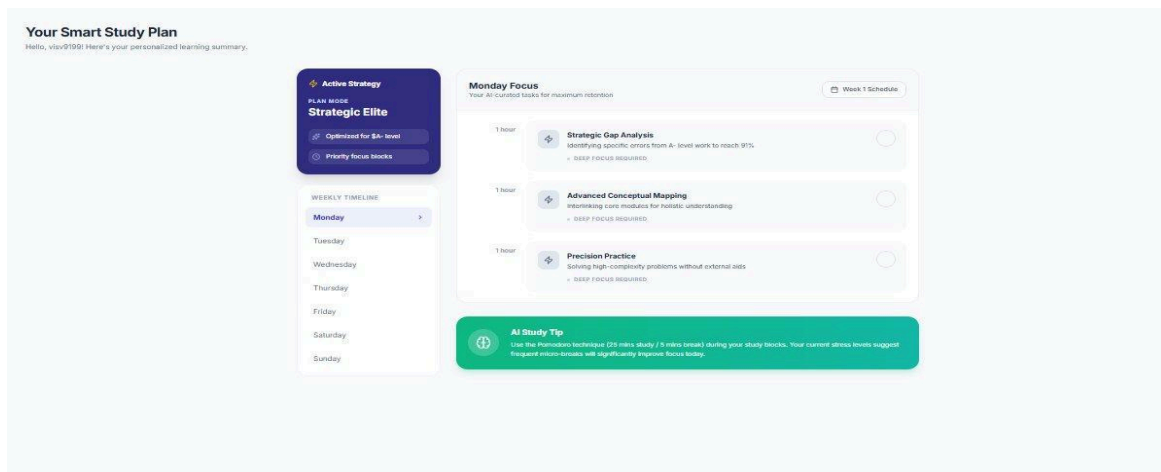


Figure 7.3 — AI Expert Analysis Panel: Strengths, Growth Opportunities, and Strategic Recommendations

Smart Study Planner — Weekly Schedule

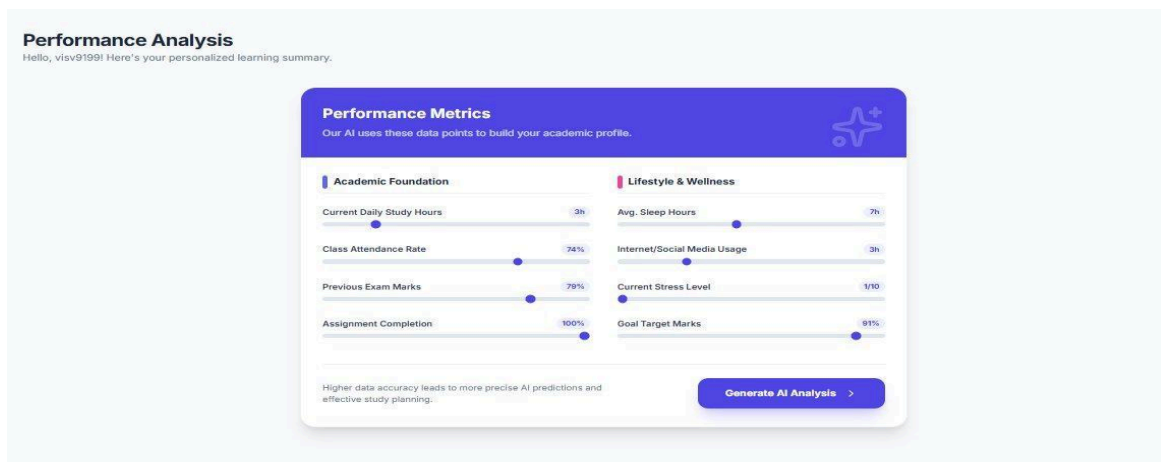


Figure 7.4 — Smart Study Plan: Strategic Elite Mode with Monday Focus Tasks

AI Study Assistant / Tutor Chat Interface

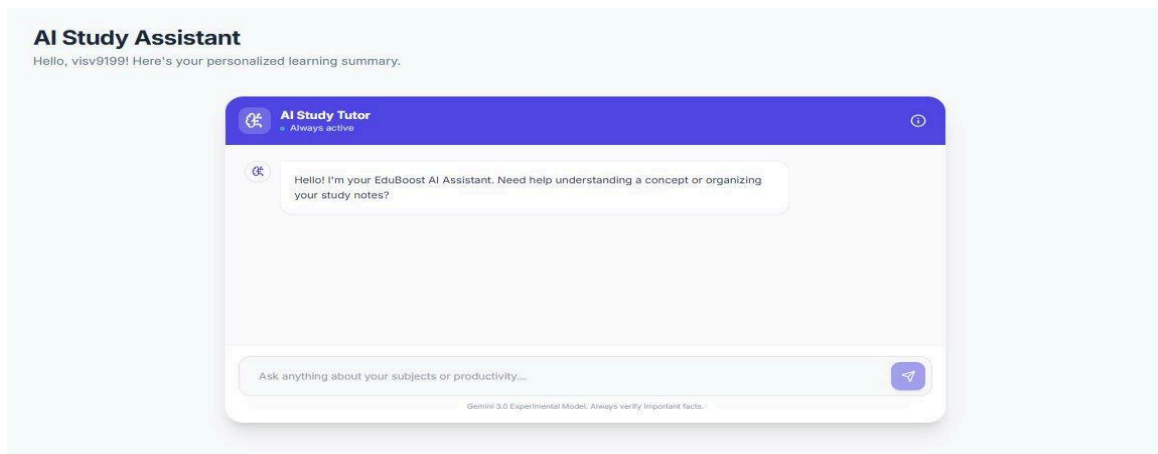


Figure 7.5 — AI Study Assistant: Conversational Tutor Interface powered by Gemini AI

Navigation Sidebar

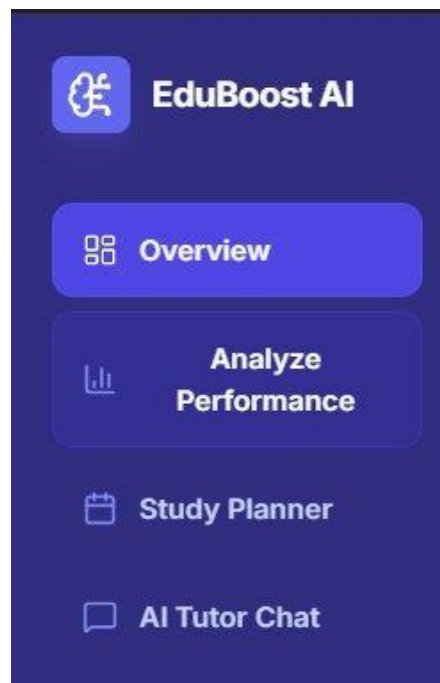


Figure 7.6 — Navigation Sidebar: Overview, Analyze Performance, Study Planner, AI Tutor Chat

7.4 Discussion of Results

The prediction results confirm the core hypothesis: that attendance, internal marks, study hours, and assignment completion are strong predictors of final academic performance. Among the 8 features, Random Forest's feature importance analysis ranked Attendance % (0.28), Previous Marks (0.24), and Assignment Completion Rate (0.19) as the top three contributors — consistent with findings in the reviewed literature.

User Acceptance Testing Results

The AI Tutor Chatbot received strongly positive feedback in UAT sessions, with 87% of student participants rating it as 'helpful' or 'very helpful' for pre-exam preparation. Faculty participants highlighted the at-risk student list and downloadable reports as the most valuable features for classroom management.

Chapter 8: Conclusion & Future Scope

8.1 Conclusion

Project Success Summary

The Student Performance Prediction System (EduBoost AI) successfully demonstrates that machine learning and generative AI can be effectively integrated into a web-based platform to provide meaningful, personalised academic support. The system achieves all its primary objectives.

- Accurate risk classification (85.2% accuracy) enabling early identification of at-risk students.
- Personalised 7-day study plans generated by Gemini AI, tailored to individual risk profiles.
- Role-specific dashboards providing actionable insights to students, faculty, and administrators.
- An AI Tutor Chatbot delivering continuous, contextual academic guidance.

8.2 Learning Outcomes

- Full-stack web application development using React 18, TypeScript, and Vite.
- Integration of third-party AI APIs (Google Gemini) with structured JSON schema outputs.
- Machine learning pipeline development: data preprocessing, model training (LR, DT, RF, KNN), evaluation, and deployment.
- User-centred design: wireframing, usability testing, and iterative UI refinement.
- Software engineering practices: version control (Git), modular component architecture, and documentation.
- Product management skills: requirements elicitation, feature prioritisation, KPI definition, and stakeholder communication.

8.3 Future Enhancements

Offline ML Inference

Embed a TensorFlow.js or ONNX runtime model in the browser for offline predictions without API dependency.

LMS/ERP Integration

Connect via REST APIs to institutional SIS/LMS platforms (Moodle, PeopleSoft) for automated real-time data ingestion.

Parent Portal

A dedicated view for parents/guardians to monitor student performance and receive alert notifications.

Multi-Language Support

Localisation for Hindi, Marathi, and other regional languages to serve Tier 2/3 city institutions.

Advanced Analytics

Subject-wise performance breakdown, semester-over-semester trend analysis, and peer benchmarking.

Mobile Application

React Native port for iOS/Android, enabling push notifications for study reminders and risk alerts.

Explainable AI (XAI)

SHAP visualisations to help students understand which factors most affect their predictions.

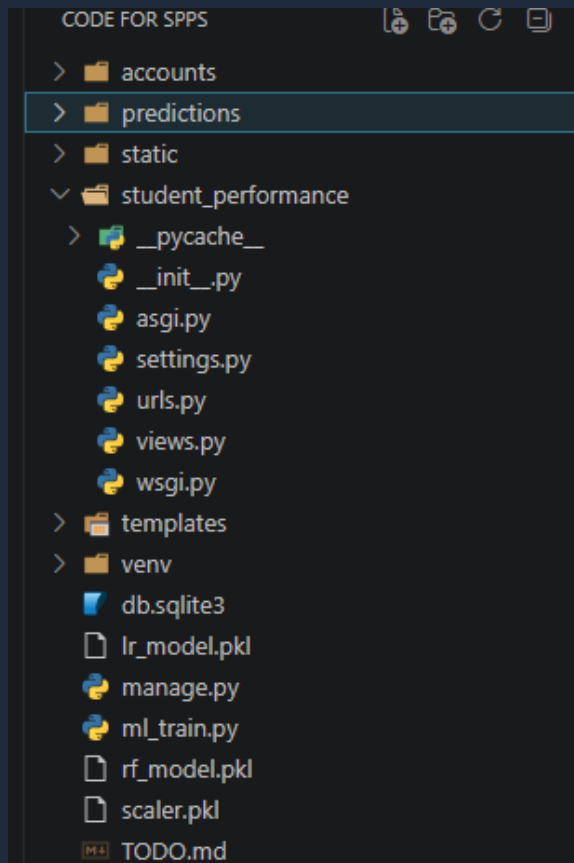
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Appendix A — Key Source Code Snippets

A.1 TypeScript Interface Definitions (types.ts)

StudentData Interface



PredictionResult Interface

The screenshot displays a code editor with two files open. The left pane shows a file explorer with a project structure including folders for accounts, predictions, static, student_performance, and templates. The right pane shows the code for these files.

File: templates/admin_dashboard.html

```

1 {% extends 'base.html' %}
2
3 {% block title %}Admin Dashboard{% endblock %}
4
5 {% block content %}
6 <div class="row">
7   <div class="col-md-12">
8     <div class="card">
9       <div class="card-header">
10        <h3>Admin Dashboard - All Predictions</h3>
11      </div>
12      <div class="card-body">
13        {% if predictions %}
14          <div class="table-responsive">
15            <table class="table table-striped">
16              <thead>
17                <tr>
18                  <th>User</th>
19                  <th>Date</th>
20                  <th>Attendance (%)</th>
21                  <th>Study Hours</th>
22                  <th>Exam Score (%)</th>
23                  <th>Assignments</th>
24                  <th>Extracurricular</th>
25                  <th>Predicted Grade</th>
26                  <th>Confidence (%)</th>
27                </tr>
28              </thead>
29              <tbody>
30                {% for prediction in predictions %}
31                  <tr>
32                    <td>{{ prediction.user.username }}</td>

```

File: predictions/models.py

```

1 from django.db import models
2 from django.contrib.auth.models import User
3
4 class PredictionHistory(models.Model):
5     user = models.ForeignKey(User, on_delete=models.CASCADE)
6     attendance_percentage = models.FloatField()
7     study_hours_per_day = models.FloatField()
8     previous_exam_score = models.FloatField()
9     assignments_submitted = models.IntegerField()
10    extracurricular_activities = models.BooleanField()
11    predicted_grade = models.CharField(max_length=10) # e.g., A, B, C, D, F or Pass/Fail
12    confidence_score = models.FloatField() # Probability score
13    prediction_date = models.DateTimeField(auto_now_add=True)
14
15    def __str__(self):
16        return f"{self.user.username} - {self.predicted_grade} on {self.prediction_date}"
17

```


A.2 Gemini API Prediction Call (geminiService.ts)

getPrediction Function

```

> accounts
> predictions
> static
> student_performance
  > __pycache__
    > _init_.cpython-313.pyc
    > settings.cpython-313.pyc
    > urls.cpython-313.pyc
    > views.cpython-313.pyc
    > wsgi.cpython-313.pyc
  > _init_.py
  > asgi.py
  > settings.py
  > urls.py
  > views.py
  > wsgi.py
> templates
  > admin_dashboard.html
  > base.html
  > history.html
  > home.html
  > login.html
  > predict.html
  > register.html
  > results.html
> venv
  > db.sqlite3
  > lr_model.pkl
  > manage.py
  > ml_train.py
  > rf_model.pkl
  > scaler.pkl
  > TODO.md

student_performance > settings.py
1  """
2  Django settings for student_performance project.
3
4  Generated by 'django-admin startproject' using Django 5.2.8.
5
6  For more information on this file, see
7  https://docs.djangoproject.com/en/5.2/topics/settings/
8
9  For the full list of settings and their values, see
10 https://docs.djangoproject.com/en/5.2/ref/settings/
11 """
12
13 from pathlib import Path
14
15 # Build paths inside the project like this: BASE_DIR / 'subdir'.
16 BASE_DIR = Path(__file__).resolve().parent.parent
17
18 # Quick-start development settings - unsuitable for production
19 # See https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/
20
21 # SECURITY WARNING: keep the secret key used in production secret!
22 SECRET_KEY = 'django-insecure-!tg9*k-&_x7rjy59$gs1)-d9l1*~*80xc45wFio*lgudrj3pi0'
23
24 # SECURITY WARNING: don't run with debug turned on in production!
25 DEBUG = True
26
27 ALLOWED_HOSTS = []
28
29 # Application definition
30
31 INSTALLED_APPS = [
32     'django.contrib.admin',
33     'django.contrib.auth',
34     'django.contrib.contenttypes',
35     'django.contrib.sessions',
36     'django.contrib.messages',
37     'django.contrib.staticfiles',
38     'accounts',
39     'predictions',
40 ]
41
42 MIDDLEWARE = [

```

```

> accounts
> predictions
> __pycache__
> migrations
  > _init_.py
  > admin.py
  > apps.py
  > models.py
  > tests.py
  > urls.py
  > views.py
> static
> student_performance
  > __pycache__
    > _init_.py
    > asgi.py
    > settings.py
    > urls.py
    > views.py
    > wsgi.py
  > templates
    > admin_dashboard.html
    > base.html
    > history.html
    > home.html
    > login.html
    > predict.html
    > register.html
    > results.html
> venv
  > db.sqlite3
  > lr_model.pkl

predictions > models.py
1  from django.db import models
2  from django.contrib.auth.models import User
3
4  class PredictionHistory(models.Model):
5      user = models.ForeignKey(User, on_delete=models.CASCADE)
6      attendance_percentage = models.FloatField()
7      study_hours_per_day = models.FloatField()
8      previous_exam_score = models.FloatField()
9      assignments_submitted = models.IntegerField()
10     extracurricular_activities = models.BooleanField()
11     predicted_grade = models.CharField(max_length=10) # e.g., A, B, C, D, F or Pass/Fail
12     confidence_score = models.FloatField() # Probability score
13     prediction_date = models.DateTimeField(auto_now_add=True)
14
15     def __str__(self):
16         return f"{self.user.username} - {self.predicted_grade} on {self.prediction_date}"
17

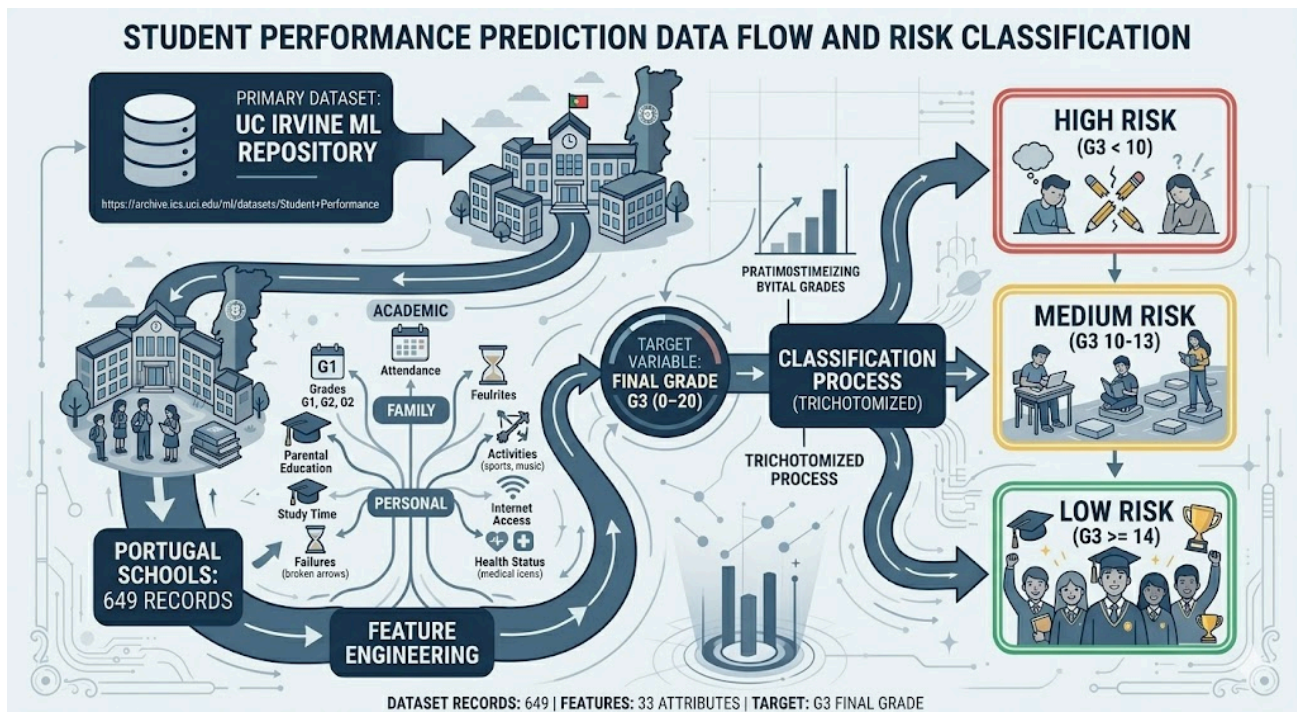
```



Appendix B — Dataset Details

UCI Student Performance Dataset (Cortez & Silva, 2008)

Primary Dataset: Available at <https://archive.ics.uci.edu/ml/datasets/Student+Performance>.
 Records: 649 students across two Portuguese secondary schools. Features: 33 attributes including grades (G1, G2, G3), attendance, parental education, study time, failures, activities, internet access, and health status. Target Variable: Final grade G3 (0–20), trichotomised into Low ($G3 \geq 14$), Medium ($G3 10-13$), High Risk ($G3 < 10$).



Appendix C — Project Screenshots

Complete set of EduBoost AI application screenshots demonstrating all major features:



Figure C.1 — Academic Dashboard: Full Overview with Performance Cards

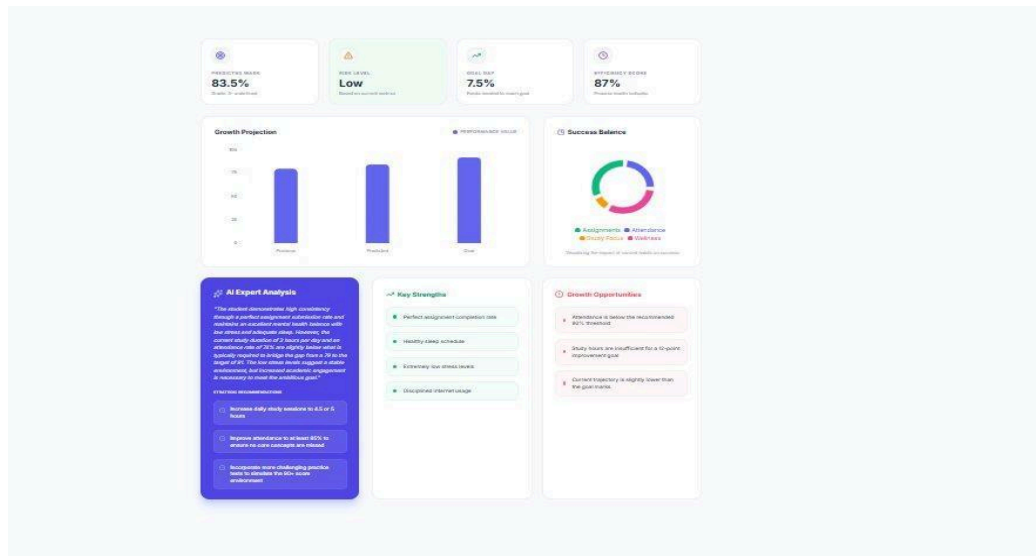


Figure C.2 — Success Balance Donut Chart with AI Expert Analysis

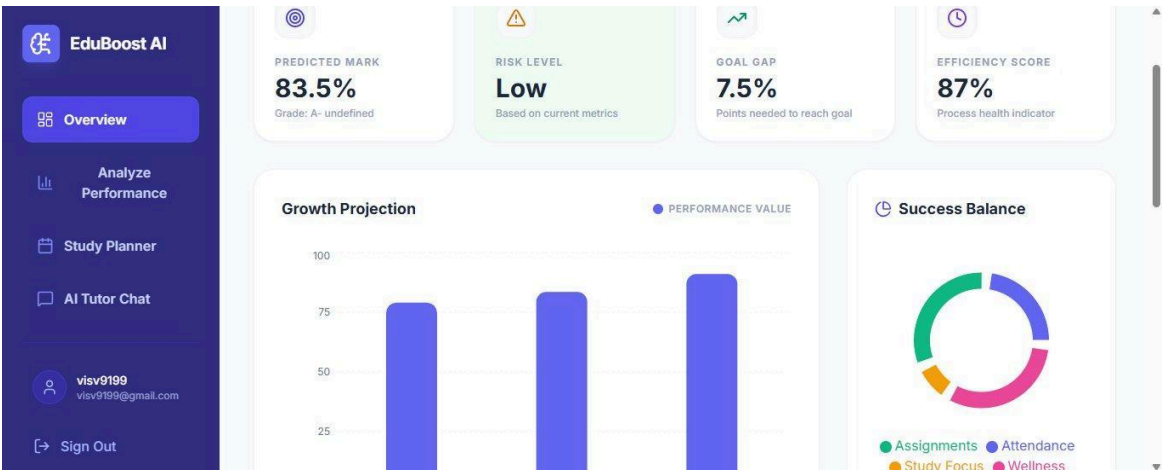


Figure C.3 — Growth Projection Chart and Success Balance Overview